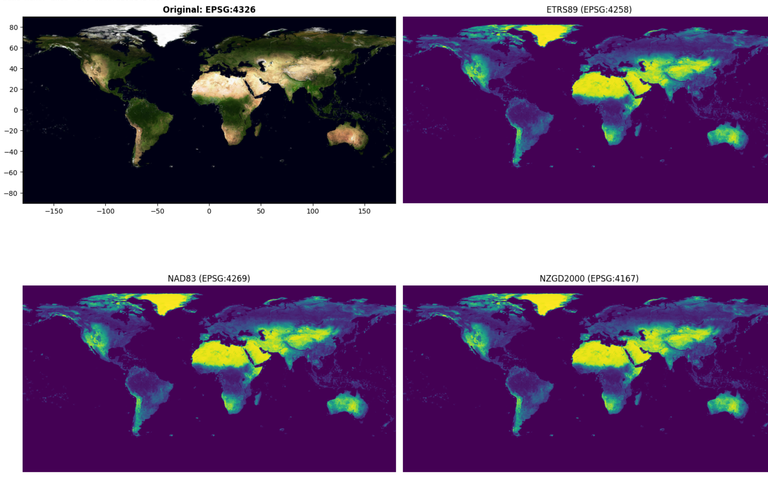
Task 9.1

In [8]:
```
import rasterio
from rasterio.plot import show
from rasterio.warp import calculate_default_transform, reproject, Resampling
import matplotlib.pyplot as plt
import numpy as np
from pyproj import CRS

with rasterio.open('earth.tif') as src:
    print(f"CRS: {src.crs}")
    print(f"\nFull CRS details:\n{src.crs.to_wkt()}")

    crs_obj = CRS.from_wkt(src.crs.to_wkt())
    ellipsoid = crs_obj.ellipsoid
    print(f"Semi-major axis (a): {ellipsoid.semi_major_metre} m")
    print(f"Semi-minor axis (b): {ellipsoid.semi_minor_metre} m")

target_crs_list = [
    ('ETRS89', 'EPSG:4258'),
    ('NAD83', 'EPSG:4269'),
    ('NZGD2000', 'EPSG:4167')
]

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
axes = axes.flatten()

with rasterio.open('earth.tif') as src:
    show(src, ax=axes[0], title=f'Original: {src.crs}')

for idx, (name, epsg) in enumerate(target_crs_list, start=1):
    with rasterio.open('earth.tif') as src:
        transform, width, height = calculate_default_transform(
            src.crs, epsg, src.width, src.height, *src.bounds
        )

        kwargs = src.meta.copy()
        kwargs.update({
            'crs': epsg,
            'transform': transform,
            'width': width,
            'height': height
        })

        destination = np.zeros((height, width), dtype=src.dtypes[0])
        reproject(
            source=rasterio.band(src, 1),
            destination=destination,
            src_transform=src.transform,
            src_crs=src.crs,
            dst_transform=transform,
            dst_crs=epsg,
            resampling=Resampling.nearest
        )

        axes[idx].imshow(destination, cmap='viridis')
        axes[idx].set_title(f'{name} ({epsg})')
        axes[idx].axis('off')

plt.tight_layout()
plt.savefig('task_9_1_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
```

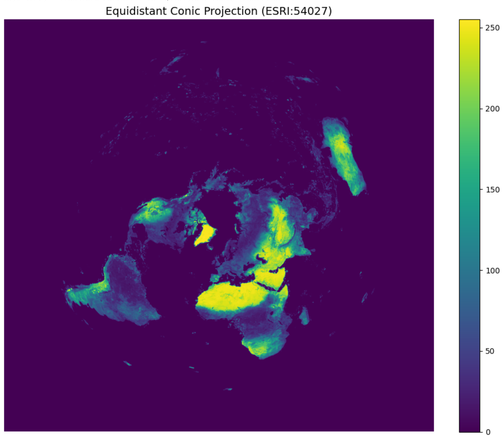
```
CRS: EPSG:4326

Full CRS details:
GEOGCS["WGS 84",DATUM["WGS_1984",SPHEROID["WGS 84",6378137,298.257223563,AUTHORITY["EPSG","7030"]],AUTHORITY["EPSG","6326"]],PRIMEM["Greenwich",0,AUTHORITY["EPSG","8901"]],UNIT["
degree",0.0174532925199433,AUTHORITY["EPSG","9122"]],AXIS["Latitude",NORTH],AXIS["Longitude",EAST],AUTHORITY["EPSG","4326"]]
Semi-major axis (a): 6378137.0 m
Semi-minor axis (b): 6356752.314245179 m
```



Task 9.2

In [9]:
```
import numpy as np
import rasterio
from rasterio.warp import calculate_default_transform, reproject, Resampling
import matplotlib.pyplot as plt

lat = 49.783333
lon = 9.933333
R = 6378137

lat_rad = np.radians(lat)
lon_rad = np.radians(lon)

print("Task 9.2.1 - Equidistant Conic Projection (origin at North Pole):")
print(f"Point: {lat}°N, {lon}°E")

rho = R * np.radians(90 - lat)
theta = lon_rad

x_conic = rho * np.sin(theta)
y_conic = -rho * np.cos(theta)

print(f"Projected coordinates:")
print(f"x = {x_conic:.2f} m")
print(f"y = {y_conic:.2f} m")
print(f"Distance from North Pole: {rho:.2f} m")

print("\nTask 9.2.2 - Centred Cylindrical Projection:")
print(f"Point: {lat}°N, {lon}°E")

phi_0_cyl = 0
x_cyl = R * lon_rad * np.cos(phi_0_cyl)
y_cyl = R * lat_rad

print(f"Projected coordinates:")
print(f"x = {x_cyl:.2f} m")
print(f"y = {y_cyl:.2f} m")

print("\nTask 9.2.3 - Transform earth.tif to Equidistant Conic (ESRI:54027):")

with rasterio.open('earth.tif') as src:
    target_crs = 'ESRI:54027'

    transform, width, height = calculate_default_transform(
        src.crs, target_crs, src.width, src.height, *src.bounds
    )

    kwargs = src.meta.copy()
    kwargs.update({
        'crs': target_crs,
        'transform': transform,
        'width': width,
        'height': height
    })

    destination = np.zeros((height, width), dtype=src.dtypes[0])
    reproject(
        source=rasterio.band(src, 1),
        destination=destination,
        src_transform=src.transform,
        src_crs=src.crs,
        dst_transform=transform,
        dst_crs=target_crs,
        resampling=Resampling.nearest
    )

    fig, ax = plt.subplots(figsize=(12, 8))
    im = ax.imshow(destination, cmap='viridis')
    ax.set_title('Equidistant Conic Projection (ESRI:54027)', fontsize=14)
    ax.axis('off')
    plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
    plt.tight_layout()
    plt.savefig('task_9_2_equidistant_conic.png', dpi=150, bbox_inches='tight')
    plt.show()
```

```
Task 9.2.1 - Equidistant Conic Projection (origin at North Pole):
Point: 49.783333°N, 9.933333°E
Projected coordinates:
x = 772274.01 m
y = -4409786.31 m
Distance from North Pole: 4476898.89 m

Task 9.2.2 - Centred Cylindrical Projection:
Point: 49.783333°N, 9.933333°E
Projected coordinates:
x = 1105773.57 m
y = 5541855.28 m

Task 9.2.3 - Transform earth.tif to Equidistant Conic (ESRI:54027):
```



Task 9.3

1. Mercator projection
2. Stereographic projection
3. Sinusoidal projection
4. Gnomonic projection
5. Equidistant Conic projection